The true alchemists do not change lead into gold;
they change the world into words.

(William H. Gass)

NLP Workbench

Throughout the NLP lecture, you will work with lots of textual data. For this reason, you will build in this warmup exercise your personal "NLP Workbench", which you can use to analyze and explore a corpus of textual data.

We will be using a dataset from Kaggle for hate speech classification:

https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge/data

The dataset consists of Wikipedia comments, labelled with the following classes: 'toxic', 'severe_toxic', 'obscene', 'threat', 'insult', 'identity_hate', or none.

Step 1: Single Texts

In the first stage, implement the core of your workbench for *a single text*. Here are some functionalities your system might provide:

- Number of characters
- Character set and number of occurrences for each character
- Full text search for a substring
- Split the text into words (using e.g. split() or nltk.Tokenizer)
- Number of words, number of different words
- An inverted index that, for given word, outputs the positions of the word in the text

Make sure to test your functions thoroughly, since they will build the core of your workbench!

Step 2: Corpus Analysis

Next, extend your workbench to handle the *entire corpus*, with functionalities such as:

- Statistics of the entire corpus about number of files, text lengths, vocabulary etc.
- Good visualizations that help to "understand" the corpus easily
- Full text search on the entire corpus for single words (using an inverted index)

Step 3: Extensions (optional)

If you want, you can extend your workbench with new, meaningful, exciting functionality. What exactly you implement is up to you. Here are some suggestions for what you could do:

- Number of sentences

- Exact search for several words linked by AND or OR
- Exact search for two words, where the first word must occur, and the second word must NOT occur
- Filter your statistics by simple criteria such as text length, occurrence of specific words etc.
- Approximative search, which also considers certain spelling mistakes (-> Levenshtein distance)
- Search with regular expressions
- Search for incomplete words (e.g. "student" also returns "students" etc. -> stemming or lemmatization)
- List most important words (-> Keyword Extraction, TF-IDF)
- A UI where you can explore the corpus interactively
- Most important topics (-> Topic Modeling)

Team Size: 1

Use of Generative AI: allowed

Hand-In: none – discussion during exercise group

Grading: not graded

Deadline: Fr 27.2.2026